Interrupting the Loop: Where Creativity Lives in a Language Model

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Abstract

Where does novelty come from when a language model writes with no task? We test three hypotheses with one measurement stack: base models run locally, cross-family LLM judges with k-sample medians and measured resolution, bootstrap confidence intervals, and, for one arm, verbatim novelty against the model's own public training corpus. **(1) The sampler.** An entropy-banded anti-probable decoder that pushes each step toward semantically distant tokens roughly doubles 4-gram novelty over min-p sampling (21.5%→45.5%, paired Δ +24.0 pp) and cuts the rate of verbatim 8+-word training blocks from 0.80 to 0.20 of generations. But this surface novelty does not rise to the level of ideas: inside a reverie loop the same push produces no measurable difference from plain sampling in judged surprise, connection or novel delta. **(2) The prompt.** Inputs composed to sit far from any human prompt do not outperform a typical request when a strong model develops them: improbable inputs are noise. **(3) The loop.** Bare continuous generation from a base model is dead: it converges on the deepest attractors of pretraining (literal orbits, website footers, exam keys) and stays there (judged surprise 0.3/10). A closed reverie loop built on the architecture of spontaneous cognition (salience-gated review, selective forgetting, reseeding, re-encounter) brings it to life (3.2; Cliff's $\delta = +0.88$, 10 seeds). Ablation and a second battery (14 conditions, 1,055 judged windows on the main generator, each judged five times) locate the effect in two operations and say what they are made of. *Habituation*, not letting the loop feed on its literal past, lifts the stream to 1.7; *interruption*, periodically injecting a new starting sentence over the preserved context, lifts it to 3.1 and carries all of the connection. The interruption must lead away (injecting the premise or the stream's own past is as bad as not interrupting), its yield depends on the thread it breaks and decays with distance (every 75 tokens is dead; the 150 tokens after breaking a 300–900-token thread score 5.3–5.9 with coherence ≈6; the best rhythm is ≈300), and salience is a good reader but a bad metronome. Inside the network (residual stream at 13 layers, logit lens), bare generation freezes at every layer, what the judge calls surprise is surface departure with deep continuity, and the interruption that works barely moves the deep state, whereas forgetting is a deep re-encounter with the beginning that the judge rewards less. The ladder bare→habituation→interruption→scaffold replicates on three generator families (Qwen3-30B-A3B, Qwen3-8B, OLMo-2-13B), under a second judge family ($\rho = 0.71$–0.85 with the first), and in a blind rating by independent human readers, whose consensus agrees with the judge ($\rho = 0.58$ on surprise) and reproduces the ordering of conditions. In this system, creativity is not in the noise injected into decoding, nor in the strangeness of the input, nor in an elaborate cognitive scaffold. It is in interrupting the loop: letting a thread develop, then making it start again somewhere else, over everything it remembers.

1 Introduction

Ask a language model for something new and you will usually get something fluent, plausible and familiar. Three places are commonly blamed, and three places are commonly optimized: the *sampler* (the tail of the next-token distribution, tamed by nucleus, typical or min-p truncation, or courted by higher temperature), the *prompt* (the input, engineered to steer the model somewhere it would not go on its own), and, less often, the *loop*, what happens when a model's output becomes its own next input with nobody asking anything.

The loop is where human novelty is usually placed. Insight rarely arrives as the answer to a strange question; it arises inside a closed circuit of thought feeding on its own output, during incubation, mind-wandering and sleep, when a spontaneously generated deviation survives critical review and is linked back to what came before. The neuroscience of creative cognition describes three coupled systems: a default-mode network that generates, an executive network that evaluates, and a salience network that decides what deserves attention; more creative people show more coupling between the first two, with salience regions coupling first [2, 3]. Creativity itself is often characterized as connecting semantically distant concepts [14], and incubation, leaving a problem and coming back to it, has a measured effect on solving it [27].

This paper asks, with controls, where novelty actually comes from in a language model left to write with no task, and which of the operations we usually credit for it survive measurement. We test the three hypotheses in turn, with one measurement stack shared by every experiment: base models (instruction-tuned models are mode-collapsed for this purpose) run locally at 8-bit; LLM judges always from a different model family than the generator, sampled k times per window with the median as the score and the intra-window spread measured before use; bootstrap confidence intervals with the window or the seed as unit; and, where a public training corpus exists, verbatim novelty computed against it.

The sampler. We build the strongest version of the "fertile error" idea we could: an entropy-banded anti-probable decoder that, at the steps where the model is genuinely undecided, rescores the plausible candidates by their semantic distance from the recent context, with a coherence floor that never admits an implausible token. It works exactly as far as the surface: 4-gram novelty against the OLMo-2 training corpus roughly doubles, verbatim 8+-word training blocks fall from 0.80 to 0.20 of generations, at a small perplexity cost. But inside a reverie loop the same push yields no measurable difference from plain sampling in judged surprise, connection or novel delta, and in verified search (bin packing, two models, 3,200 verified heuristics) it does not separate from plain sampling either. Surface novelty is real and does not climb, the mechanistic counterpart of the recent finding that most n-gram-novel expressions are not judged creative [26].

The prompt. We build inputs that sit far from any human prompt (distant concept pairs, composed improbable contexts; distance measured against 10,000 real prompts) and let a strong model develop them. A typical request matches or beats every improbable arm; input improbability does not correlate with judged novelty. Improbable inputs from outside are noise.

The loop. Bare continuous generation from a base model is dead: left to feed on its own output it converges on the deepest attractors of pretraining (literal orbits, website footers, exam keys, translation tables) and stays there. A closed reverie loop built on the architecture of spontaneous cognition (a salience monitor over the stream's own telemetry gating review, selective forgetting, reseeding, re-encounter with the premise) brings it to life. Then we take the loop apart. Ablation and a second battery of 14 conditions locate almost the whole effect in two operations, neither of them the elaborate ones: *habituation*, which keeps the loop from eating its literal past, and *interruption*, a new starting sentence injected every few hundred tokens over the preserved context. It is the machine equivalent of getting up for a coffee: leaving a thread one has been developing and being made to start somewhere else, remembering everything. We show what a good interruption is made of: it must lead away, its yield depends on the length of the thread it breaks and decays with distance from it, the best rhythm is about 300 tokens, and a salience monitor is a good reader of the stream but a bad metronome for it. We replicate the ladder on three generator families and under a second judge family. Finally we look inside the network: residual-stream geometry at 13 layers shows that bare generation freezes at every depth, that what the judge calls surprise is surface departure with deep continuity, and that the interruption that works barely moves the deep state, whereas forgetting is a deep re-encounter with the beginning that the judge rewards less.

Contributions.

1. A calibrated anti-probable decoder and the measurement that its novelty is real at the surface (objective, against the training corpus) and absent at the level of ideas (Sections 4, 7).
2. A negative result on improbable inputs, with input improbability measured (Section 5).
3. A reverie loop with a salience monitor and selective forgetting, and its ablation into two operations, habituation and interruption, with a battery that characterizes the interruption's content, timing and rhythm, replicated on three generator families (Section 6).
4. A residual-stream analysis relating judged surprise to what the network does at each depth, and a dissociation between deep representational change and judged quality (Section 8).
5. An instrument section that measures the LLM judge before using it and validates it against a second judge family and against independent human readers (Section 7).

Code, per-run data, judgments and the dated laboratory notebook are released with the paper.

2 Related work

Decoding and the tail. Nucleus sampling [13] truncates the unreliable tail of the next-token distribution; locally typical sampling [18] targets human-like surprisal; min-p [21] scales the truncation with the model's confidence and is exactly the coherence floor we use; contrastive decoding [16] and plug-and-play steering [7] shape the distribution toward or away from a reference. All of these regulate the tail. Our anti-probable sampler *courts* it, inside an entropy band and above a floor, and we find that this buys surface novelty only.

Repetition, self-reinforcement and entrainment. The collapse of a base model left to itself is a well-studied phenomenon. Zhu et al. [30] name the *self-reinforcement effect* (a repeated token becomes more probable each time it repeats) and suppress it with a repetition penalty restricted to a recent window; the graded, windowed penalty we call *habituation* is a decoding-time operator of the same family, and we claim no novelty for it. Guan and Huang [10] attribute the tendency to repeat to a learning bias of maximum-likelihood training and correct it with self-contrastive training; Xu et al. [29] track the distance between the current next-token distribution and past ones to detect repetition and topic drift and steer decoding accordingly. At the mechanistic level, Niu et al. [22] show that language models assign higher probability to tokens present in their context even when those tokens are irrelevant (*contextual entrainment*), and locate heads that carry it. Our bare arm is entrainment on the model's own output; what this paper adds is not a new remedy for repetition but a measurement of what the remedy leaves untouched (a habituated stream is fluent and still not creative) and of what a second, structural operator adds on top of it.

Evaluating creativity, and the judge. Nakajima et al. [20] show that the widely used Divergent Association Task ignores appropriateness and propose to score novelty conditional on it; our three separate dimensions (surprise, connection, coherence), reported without a composite, follow the same logic of not letting novelty be bought with incoherence. Saakyan et al. [26] show with 8,618 expert annotations that ≈91% of top-quartile n-gram-novel expressions are not judged creative; our factual-paraphrase escape mode and the sampler's idea-level null are independent confirmations of the same gap. Infini-gram [17] and Rusty-DAWG [19] make verbatim novelty against a training corpus computable at scale; we use infini-gram against the OLMo-2 corpus. On the instrument itself, Fein et al. [9] find that the strongest off-the-shelf judge they tested agrees with human preferences on creative-writing pairs only 73% of the time, and Haldar and Hockenmaier [11] document low intra-rater reliability of LLM judges across runs. Both cautions apply to this study. We answer the second by scoring every window with $k = 5$ independent calls and using the median, and by reporting the resolution of the instrument; we answer the first only partly, with a second judge family and with human raters on a subset, and we treat the judge as the study's main limitation (Section 10).

Loops and steering in story generation. Outer loops that steer a generator are common in story generation: SWAG [24] lets a second model choose the next action for the story model; Collective Critics [1] refine a plan and its expression with several critic models; STORYTELLER [15] adds a plot-planning structure to keep long stories coherent and cohesive. Against this literature our contribution is not the idea of an outer loop but its reduction: the operator we isolate is one injected sentence on a preserved context, without a planner, a critic or a reward; the factorial ablation that shows which part of the loop carries the effect; and the temporal characterization (period, decay, timing) that a planning framework does not provide.

Verified search. FunSearch [25] and AlphaEvolve [23] pair a language model with a hard evaluator and let selection do the work. Our bin-packing null (the anti-probable operator does not separate from plain sampling under verified evolution) and the loop results suggest that the structure of the generation loop, not the sampling operator, is where to look next.

Reading the residual stream. The logit lens is known to be fragile in intermediate layers, and the tuned lens [4] was proposed as a less biased alternative. We use the logit lens only for a coarse commitment layer and otherwise work with mean-centered residual geometry; the network section is descriptive, and its readings should be repeated with a tuned lens before they are taken as mechanism.

Creative cognition. Creative thought as coupling of default-mode and executive networks [2], with salience regions coupling first [3]; creativity as connecting semantically distant concepts [14]; incubation effects [27]; predictive processing [6]; dreams as anti-overfitting noise [12]; Boden's novelty/surprise/value triad [5]; conceptual blending [8]; novelty search [28]. The reverie loop of Section 6 was an explicit engineering of the first three into a text loop; its ablation is, in a sense, a test of which of these ingredients a language model needs, and the answer (an interruption on a clock) is humbler than the hypothesis.

3 Method

3.1 Measurement stack

Every experiment shares one stack. **Generators** are three base models from two families: Qwen3-8B-Base and Qwen3-30B-A3B-Base (a mixture-of-experts model with 3B active parameters, 53 tokens/s, the main generator), and OLMo-2-13B (whose training corpus is public). All are quantized to 8 bits (MLX affine quantization, group size 64) and run on Apple silicon through MLX with a numpy sampler that sees the full logits. Restricting to base models is a scope decision, not a finding: an instruction-tuned model brings a dialogue format and preference training that change what an open-ended continuation is, and whether the operators studied here transfer to it is left open (Section 10). **Judges** are Claude Opus 5 and, in the instrument section, Claude Sonnet 5 (Amazon Bedrock) and Kimi K2.6 (OpenRouter). In the loop experiments the judge is always from a different family than the generator; the idea experiment of Section 5 used Claude both to develop and to judge, a within-family judgment we flag there. Each window is judged k times ($k = 5$ in the loop experiments; $k = 3$ in the earlier ones) with independent 0–10 dimensions and the median is the score. For loop windows the dimensions are *surprise* (how unexpected the window is given the earlier text; 0 = the obvious continuation, 10 = startling yet not random), *connection* (does it bring together two distant regions of the earlier text, or an old region with something new, in a way that makes sense) and *coherence* (does it hold together as text; 0 = word salad or document boilerplate); for ideas, a nearest-equivalent plus novel-delta rubric. The full judge prompts, model identifiers, parameters and dates are in Appendix A. **Instrument calibration** (Section 7) measures the intra-window spread of the k judgments and the judge's agreement with a second judge family and with human raters. **Objective novelty**, where a public corpus exists, is computed with infini-gram [17] against the OLMo-2 corpus.

Unit of analysis and statistics. A *cell* is one premise run under one condition for 4,500 generated tokens; each condition has ten cells, one per premise, and conditions share the ten premises. The unit of inference is the cell: the windows of a cell are averaged into one score per dimension, and every comparison between two conditions is a paired comparison of ten cell means. We report the mean over cells with a bootstrap confidence interval over cells, and for each pair of conditions the mean paired difference with its bootstrap CI, an exact sign-flip permutation p-value on the ten paired differences (all 2^{10} sign patterns), and Cliff's δ on the cell means. Where a question involves several comparisons we report Benjamini–Hochberg q-values within that family alongside the raw p. Window-level statistics (bootstrap on windows, Mann–Whitney) are reported only in the appendix, as descriptive, because windows from the same stream are not independent. This seed-level analysis and the window protocol below were adopted after an external review of a first version of this manuscript, which analyzed windows as the unit; the arms of battery 3 were generated after that change, the earlier arms were re-judged and re-analyzed under it.

3.2 The anti-probable sampler

At each step the model's next-token distribution is computed as usual. Tokens below a relative probability floor ($p < 0.05\, p_{\max}$, the min-p rule) are never chosen. If the entropy of the distribution falls inside a band $[H_{\min}, H_{\max}]$ (the model is undecided but not lost), the remaining candidates (at most 128) are re-scored by

$$\mathrm{score}(t) = \log P(t \mid \mathrm{context}) + \lambda \cdot z\big(d(t,\ \bar{c})\big),$$

where d is the cosine distance between the token's input embedding and an exponential moving average $\bar{c}$ of the recent context in the model's own embedding space (half-life 16 tokens in the sampler experiments, 48 in the loop), and z standardizes the distances across the step's candidates so that λ reads in nats per standard deviation; the token is then sampled from the softmax of the scores. Below the band the model is confident and we let it be; above it the distribution is a genre fork and pushing there produces collapse. The band was $H_{\min} = 2.0$, $H_{\max} = 4.5$ nats in the sampler experiments and $[1.8, 4.5]$ in the loop's drift regime; the values

Premise: *Every morning the baker counted the loaves twice, and every morning the count was different.*

...the stream, 2,000 tokens later: *There was never more than thirty, but there were always twenty-one or twenty-two. That morning he counted them three times, but he got no nearer to the number. He couldn't understand it. He opened the door of the oven and pulled out a loaf of bread. He cut it open. It was all*

Injected: *Meanwhile, in a city with no name,*

The model continues: *an old woman sat at a window with her cat. She had no idea who she was or where she came from, or why she'd been left alone in this strange city with this strange cat. All she knew was that every day something went wrong with the cat: It would grow ill for no reason and then recover; it would lose its fur for days; even die for a few hours! Every day this happened, but every day it also recovered. And then one day, when the old woman looked out...*

Figure 1: A real interruption (Qwen3-30B-A3B, seed 2, period 300). The injected sentence is indistinguishable from the model's own text; the baker remains in the context and can be returned to.

were set on calibration probes of Qwen3-8B and transferred unchanged to the other models. Optional terms used in the loop are a *habituation* penalty (the probability of a token that occurred in the last 512 positions is divided by 1.15^m, where m is the number of its occurrences in that window, before the floor and the re-scoring) and a *bridge* bonus (weight 1.5 in the full scaffold) that rewards candidates close to anchor regions visited long ago and far from the recent context.

3.3 The reverie loop

Forced continuation without a task. A cell starts from a single sentence, the *premise* (e.g. *"The town had two clocks, and nobody remembered which one had been right first."*), and nothing else: no instruction, no role, no question. A base model continues it, and from then on its only input is its own output: each token is predicted from the premise plus everything the model has written. In all arms but one the end-of-text token is masked, so the stream never stops by itself; the model is forced to keep writing one document for 4,500 tokens. This is a deliberately hard regime, and part of what the paper measures is how a base model behaves in it.

Interruption. At chosen moments the generator is paused and a short text is *injected* into the stream as if the model had written it (pre-filled into the model's cache), and generation resumes from there. The model receives no signal that the text was injected: to it these are the next tokens of the document. Everything written before stays in the context: the model keeps all of it, and is only made to open a new sentence somewhere else. Figure 1 shows a real interruption. The default injected texts are four neutral subject changes used in rotation (*"That night she dreamed of something else entirely:"*, *"Meanwhile, in a city with no name,"*, *"There is an older story about this, and it goes:"*, *"A question nobody had asked yet:"*); other contents are studied in Section 6.

The full scaffold (DREAM). The architecture we started from adds, around the same stream: a *salience monitor* over the stream's own telemetry (a fast moving average of token embeddings and the step entropy), which fires on a semantic *jump* (the average moved far from where it was 32 tokens earlier), a *crystallization* (entropy dropped after a stretch of wandering), a *recurrence* (the average came close to a region visited long ago and not recently), and *stagnation* (it has not moved for a long stretch); a surface *genre-collapse* detector (lexical diversity, capitalization, layout symbols); a judge (Claude Sonnet 5, $k = 3$) called on salience events during generation, whose passing verdict (median score ≥ 5) triggers *escalation* (a narrower regime plus an injected textual return to the premise); a *kick* on stagnation (a stretch of harder push), and after repeated stagnation a subject change with *selective forgetting*: the working memory is rebuilt in a fresh cache from the premise, up to three judged-good windows and the last 200 tokens, plus the new seed, so that a deep well cannot pull the seed back; regions visited persist as anchors for the bridge term. The full scaffold is the only arm in which a judge is called inside the loop. In the main generator's scaffold cells the judge was on and reviewed 80 salience events; none reached the threshold, so escalation and the judged re-encounter never fired and the judge changed no generated token; in the replications on the other two generators the scaffold ran without the in-loop judge. In practice, then, the scaffold's active mechanisms were the kick and the reseed with forgetting, and in every arm of this paper the scores come from a judge that sees the stream only after generation. Every one of these mechanisms was born from a calibration probe in which a base model without a task fell into a specific degeneration mode (early end-of-text, erudite rumination, literal four-sentence orbits, website footers, translation tables). The generator's pretraining diet was the dominant variable in those probes: OLMo-2 sinks into footers even with forgetting; Qwen3-30B drifts in prose. With 3,000 tokens of boilerplate in the cache, every injected seed was pulled back within about twenty tokens: the accumulated context is the well, and forgetting is what lets a seed take. The results below show that this architecture is not what carries the effect; it is kept as the origin of the study and as one arm.

Conditions. Table 1 lists the arms. All loop arms use $\lambda = 0$ (no anti-probable push), since the push had no detectable effect inside the loop (Section 4); the arms differ in whether the habituation penalty is on, in whether the end-of-text token is masked, in whether the context is preserved or reset at an interruption, and in whether, when, how often and with what the stream is interrupted. Ten narrative premises (Appendix A) are shared by all arms; the sampler's random-number seed is the same (0) in every cell, so two arms that make identical decisions up to a point produce identical tokens up to that point, and the premise is the only source of variation between the cells of an arm.

Judging the stream: generated-only windows. After generation, each cell is scored on windows made only of model-generated tokens. In an interrupted cell, a window of 96 tokens starts 32 tokens after the end of each injected sentence (the *post-interruption window*); further windows start 160, 300, 450, 600 and 750 tokens after the injection when the segment is long enough, and no window crosses an injection. In a cell without injections the same window shape is cut on a uniform grid every 150 tokens. The judge sees the 600 tokens preceding the window as context (which may contain an injected sentence) and grades only the window, on the three dimensions, five times; the median is the window's score. Up to six post-interruption windows per cell, evenly spread over the stream, and three per deeper offset are judged. The primary measure of a cell is the mean of its post-interruption windows (grid windows in uninterrupted arms); the deeper offsets give the decay curve of Section 6. Each cell stores its token stream and the exact position of every event and injection. The protocol of the first version of this study, which cut 160-token windows right before each injection and on a 150-token grid so that a window could contain an injected sentence, is kept in Appendix C for comparison.

4 Two motivating nulls: the sampler and the prompt

The program began where effort is usually spent, the sampling operator and the input, and found nothing at the level of ideas in either place. Both studies are small and are reported here as motivation for the loop experiments, with their limits stated; the full tables are in the repository.

arm	habituation	interruption
bare	off	none: continuous generation, EOS masked
bare + habituation	on	none
habituation, EOS allowed	on	none; the model may emit end-of-text and start a new document
habituation 1.3	on (stronger)	none
interruption, no habituation	off	neutral subject change every 150 or 300 tokens, context preserved
habituation + interruption 150	on	neutral subject change every 150 tokens, context preserved
salience only	on	none; salience events mark review windows
DREAM scaffold	on	full scaffold: salience, in-loop judge, kick, reseed with forgetting, re-encounter
content: re-encounter / premise / own past	on	every 150 tokens: a return-to-the-premise stitch / the premise itself / a window of the stream's own past ($\geq$400 tokens back)
salience: re-encounter	on	the stitch injected on each salience event, no judge gate
period 75 / 300 / 600 / 900	on	neutral subject change at other periods (900 also with the stitch)
sham break 300	on	a paragraph break ("\n\n") every 300 tokens, nothing else
sham continuity 300	on	"And so, as before," every 300 tokens: a boundary that asks for continuity
reset + subject change 300	on	neutral subject change every 300 tokens on a *reset* context (premise + injected sentence only)
reset + break 300	on	a paragraph break every 300 tokens on a reset context

Table 1: Conditions: the habituation × interruption factorial with its baselines (top), the content and timing arms (middle) and the boundary and context controls (bottom). Ten premises each, 4,500 tokens per cell, main generator Qwen3-30B-A3B-Base; the ladder bare / habituation / habituation + interruption 150 / scaffold is replicated on Qwen3-8B-Base and OLMo-2-13B.

The sampler doubles surface novelty. On OLMo-2-13B, whose training corpus is public, we ran a fully paired grid of 3 prompts × 5 seeds × {min-p baseline, $\lambda = 1$, $\lambda = 2$} and counted, with infini-gram, which n-grams of each generation occur anywhere in the corpus (Figure 2). Novel 4-grams rise from 21.5% (baseline) to 36.0% ($\lambda = 1$) and 45.5% ($\lambda = 2$): paired bootstrap Δ +14.6 pp $[7.9, 21.9]$ and +24.0 pp $[18.2, 29.5]$, monotone in λ and present in each prompt separately. The share of generations containing a verbatim training block of eight or more words falls from 0.80 to 0.20 in both machine arms. The coherence cost is about +1.2 perplexity under a cross-family model, uncorrelated with novelty within arms (Spearman +0.12). Three escape modes were mapped (recitation, collage and factual paraphrase), and the entropy ceiling of the band, above which the model is at a genre fork and pushing produces collapse, transferred across model families.

And has no detectable effect at the level of ideas. Two tests. In verified search (online bin packing, the FunSearch benchmark, with a deterministic verifier in place of a judge; evolutionary loops of 5 runs × 8 generations × 20–40 candidates on Qwen3-8B and Qwen3-30B-A3B, 3,200 verified heuristics on the larger model) the anti-probable operator does not separate from plain sampling: both arms rediscover best-fit at will and neither beats it on held-out instances. And inside the reverie loop of Section 6, the same push against plain sampling in the same scaffold shows no detectable difference in judged surprise, connection or novel delta across 1,335

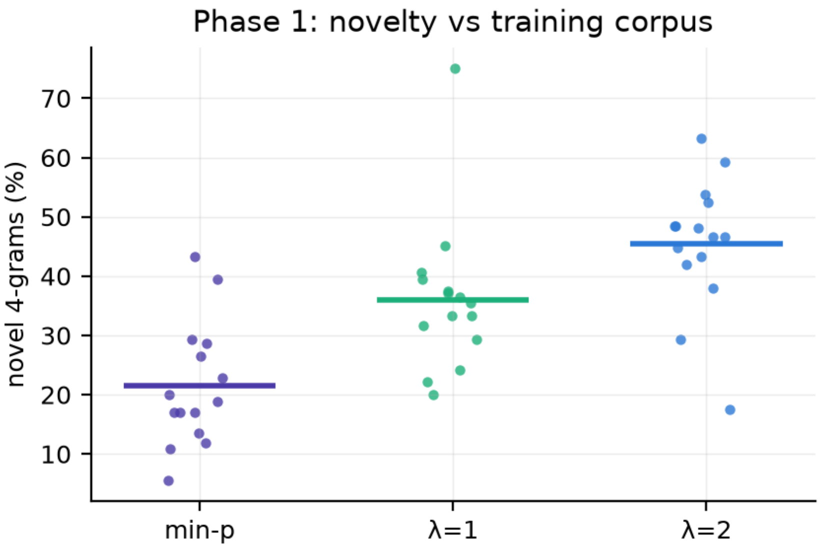

Figure 2: Objective novelty against the OLMo-2 training corpus: novel 4-grams per cell, 3 prompts × 5 seeds. The anti-probable push roughly doubles surface novelty over min-p sampling.

window-level judgments by two judges under three rubrics (the surprise rubric: 2.97 vs 3.48, CI of the difference [-1.50, +0.49]). With ten cells per arm and a judge whose per-window noise is about ±0.7, an effect of half a point would not have been detected; the conclusion is a null at the present resolution, not an absence.

5 The prompt

We composed three kinds of input (a typical request, a distant concept pair drawn from a band of embedding distances, and a composed improbable context) plus two ablations (fragments only, register only), had Claude Opus 5 develop each into an idea, and had Claude Sonnet 5 judge the result against its nearest known equivalent ($k = 3$; $n \approx 15$ per arm; developer and judge from the same family, the one within-family judgment in this paper). Input improbability was measured two ways: semantic kNN distance to 10,000 real instructions (Alpaca, embedded with all-mpnet-base-v2) and perplexity. The typical prompt matched or beat every improbable arm, two of which scored lower by a margin whose interval excluded zero; improbability did not correlate with judged novelty ($|r| < 0.2$, Figure 3); a pilot effect ($n = 8$) had not replicated. We found no benefit from prompt improbability under the tested operationalizations; whether other operationalizations, developers or judges would find one is open.

6 Hypothesis 3: the loop

All results in this section use Qwen3-30B-A3B-Base unless stated, ten narrative seeds, 4,500 tokens per cell, $\lambda = 0$, and Claude Opus 5 as judge with $k = 5$. Full tables, including Cliff's δ, Mann–Whitney p-values and paired-by-seed intervals, are in Appendix D.

6.1 Bare generation is dead; the loop brings it to life

Left to continue a premise with nothing else, the base model converges within a few hundred tokens on an attractor of its pretraining and stays there: *"Highly recommended. Highly recommended. . . . "* for two thousand tokens; a table of three-digit numbers; an English-exam answer

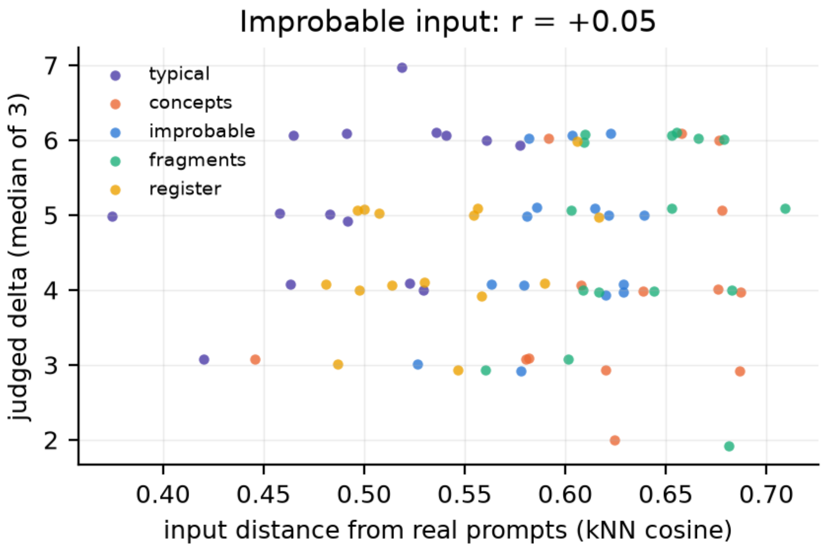

Figure 3: Judged novel delta of the developed idea against the input's semantic distance from 10,000 real prompts, per arm. No relationship.

key. Judged surprise 0.28, connection 0.20, coherence 2.06 (50 windows). The full scaffold brings the stream to 3.19 / 1.64 / 3.68 (Cliff's $\delta = +0.88$ vs bare on surprise, $p \approx 5 \times 10^{-15}$; every seed moves in the same direction, Figure 4), and elaborates its own finds after a re-encounter (*"Everything that almost happened piled up inside her like snow falling sideways through open windowsills into rooms filled with silence instead of noise . . . She kept a notebook because sometimes almosts piled up too much to be ignored"*). Salience-gated review also beats reviewing on a timer as a policy for *which* windows to judge (surprise and connection CIs positive at a third of the judge cost).

6.2 Taking the loop apart

Ablation over 10 seeds (Figure 5, Table 2) gave the first surprise: *bare + clock reseed* (no salience, no forgetting, no kicks, only a neutral subject change every 150 tokens over the preserved context) ties the full scaffold on surprise (3.08 vs 3.19, CI [-0.54, +0.78]) and beats it on connection (3.15 vs 1.64, $\delta = -0.58$ for the scaffold, $p \approx 1.5 \times 10^{-7}$) and coherence (4.78 vs 3.68), with 12/60 windows judged both surprising and coherent against 8/47 for the scaffold and 0/50 for bare. Salience-only (review windows marked but nothing injected) reaches 2.63. In sentence-embedding space (Figure 6) bare generation has explored radius 0.18 and mean step 0.14 and ends 0.78 from the premise; the scaffold has radius 0.51 and returns to within 0.57 of the premise; the clock reseed spreads over the space in jumps while keeping the whole context, which is where its connection advantage comes from.

6.3 What the interruption is made of

The ablation left the effect in one operation and raised four questions about it, and one about itself. Battery 2 answers them (Table 2, Figures 7–10).

A confound, and two operations. The bare arm of the ablation ran without the scaffold's habituation penalty, which every interrupted arm had; bare generation dies in *literal* orbits, precisely what a repetition penalty undoes. Adding the missing control (habituation, no interruption) splits the effect in two: bare 0.28 → bare + habituation **1.70** (surprise Δ +1.42

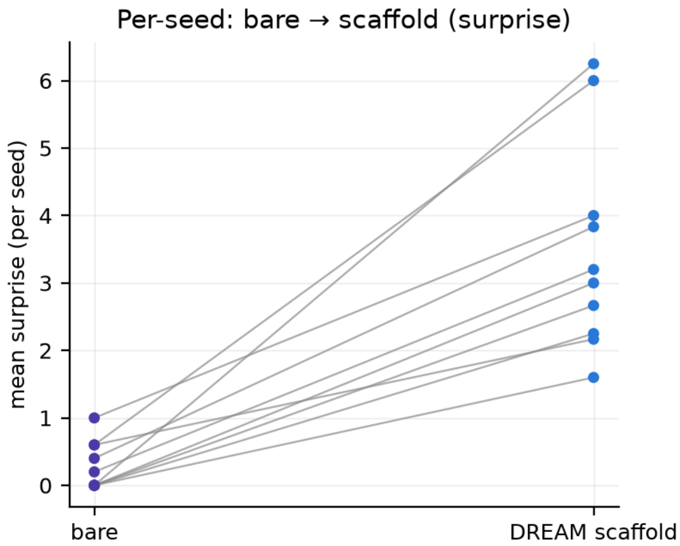

Figure 4: Per-seed robustness: mean surprise of bare vs scaffold windows, one line per seed.

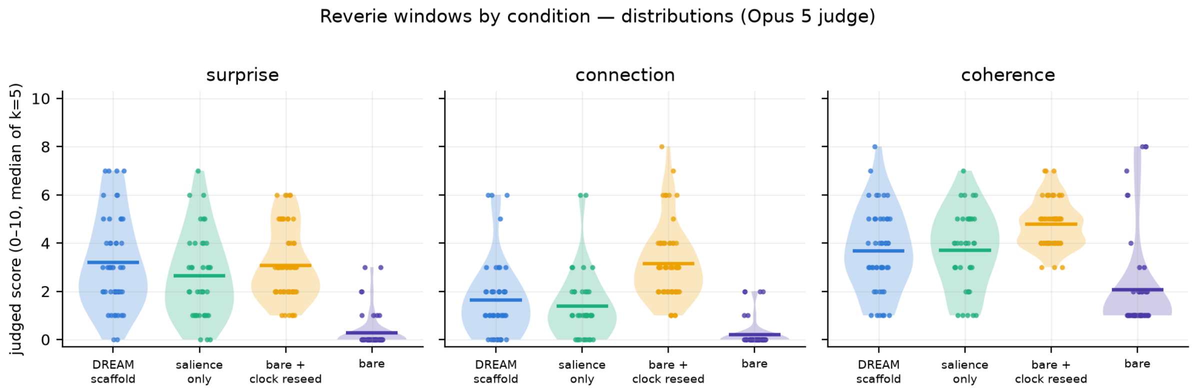

Figure 5: Ablation: judged surprise, connection and coherence by arm (10 seeds, Opus 5, median of $k = 5$ per window; violins are distributions, bars are means with bootstrap CIs). Bare + clock reseed matches the full scaffold on surprise and exceeds it on connection and coherence.

$[+1.12, +1.72]$ paired by seed; connection $0.20 \to 1.08$; coherence $2.06 \to 4.08$) $\to$ bare + habituation + clock reseed **3.08** (Δ +1.38 $[+0.92, +1.92]$). Habituation removes the literal orbits and buys about half of the surprise; the interruption buys the other half and *all* of the connection ($1.08 \to 3.15$, Δ $+2.07$ $[+1.63, +2.57]$, $\delta = +0.75$) and the coherence. The full scaffold adds surprise +1.49 $[+0.80, +2.19]$ over habituation alone but only +0.55 of connection.

The interruption must lead away. With the same period (150), a neutral subject change and a re-encounter stitch that asks the stream to return to its beginning are close (3.08 / 3.15 / 4.78 vs 2.68 / 3.64 / 4.72; the stitch trades some surprise for connection, not significantly). Injecting the *premise itself* (1.14 / 1.04 / 3.18) or a window of the stream's *own past* (1.42 / 1.52 / 3.06) is as bad as not interrupting at all: every paired CI against the neutral change excludes zero (connection -2.11 [-2.83, -1.46] and -1.63 [-2.37, -0.91]; $\delta \approx -0.6$ to -0.7). Qualitatively, the own-past injection reinforces whatever attractor the stream is in: an exam-key well was fed its own questions back. A return works only as a short stitch that still opens a new sentence; a literal return closes the loop (Figure 8).

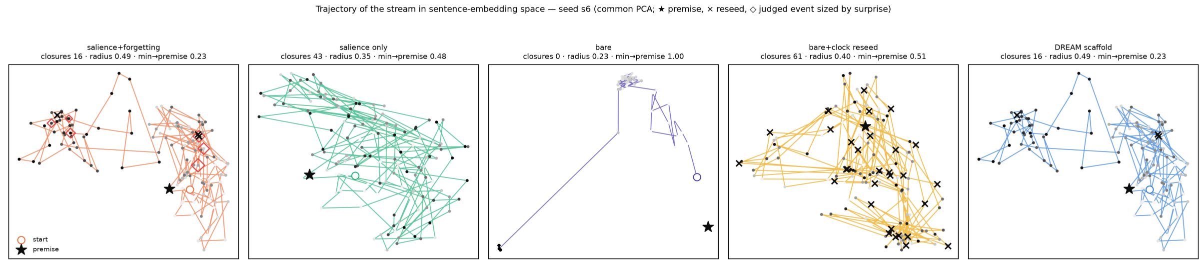

Figure 6: Where the stream goes: trajectories in sentence-embedding space (64-token windows, one PCA per seed), seed 6. ⋆ premise, ∘ start, × reseed, ⋄ judged event sized by surprise. Bare makes one jump and freezes; the scaffold orbits and returns; the clock reseed jumps across the space keeping its context.

	Qwen3-30B-A3B			Qwen3-8B			OLMo-2-13B		
arm	S	C	H	S	C	H	S	C	H
bare	0.28	0.20	2.06	0.68	0.38	3.45	1.35	0.91	2.97
bare + habituation	1.70	1.08	4.08	1.47	0.87	3.93	2.37	1.51	4.05
bare + habituation + clock reseed	**3.08**	**3.15**	**4.78**	**2.73**	**3.23**	4.42	**3.11**	**3.17**	3.28
DREAM scaffold	3.19	1.64	3.68	2.73	1.53	4.30	3.08	1.89	**4.48**

Table 2: The ladder on three generator families: judged surprise (S), connection (C) and coherence (H), means over windows (Opus 5, $k = 5$; 10 seeds per arm; windows before 100 generated tokens excluded). Habituation buys part of the surprise; the interruption buys the rest and all of the connection; the scaffold adds nothing over the plain interruption except, on OLMo, coherence (its selective forgetting lifts the stream out of a web-boilerplate well).

Salience is a good reader and a bad metronome. The stitch injected on salience events (jump, crystallization, recurrence; 1–9 per cell, median ≈5) makes the *event windows* better than salience-only (3.92 / 2.99 / 3.75 vs 2.63 / 1.39 / 3.71), but the *stream* it produces is poor: uniform windows 2.18 / 1.68 / 3.93 against **4.70 / 3.48 / 5.50** for the same stitch on a clock of matched frequency (every 900 tokens; surprise Δ +2.02 [+1.62, +2.44] paired, coherence +0.78) and 5.48 / 2.87 / 5.95 for a neutral change every 900 (45/60 windows both surprising and coherent, the best cell of the program). Salience events select good moments to look at, which is why salience-gated review beat clock review; as an interruption trigger they fire rarely and unevenly and leave streams stuck for thousands of tokens. Salience should decide where to look, not when to interrupt (Figure 9).

The yield of an interruption depends on the thread it breaks, and decays with distance from it. With the neutral change every 75 tokens the stream is dead (0.88, $\delta = -0.83$ vs 150): nothing has time to develop. Every 150: 3.08. Every 300, 600 and 900, the windows that end within ≈160 tokens after an interruption score **5.28 / 3.35 / 5.95**, 5.40 / 3.20 / 5.65 and 5.90 / 2.80 / 6.10, the highest values in the program (34/60 of the 300-early windows both surprising and coherent), while windows deeper into a segment fall back to ≈3.2 and stay there. The same window shape (a break plus 150 tokens of development) scores 3.08 when the broken thread was 150 tokens long and 5.3–5.9 when it was 300–900: surprise needs a developed background to break. Weighting the two phases by their share of the stream, period 300 gives the best stream (4.35 / 3.18 / 5.97), then 600 (3.82), 900 (3.68), 150 (3.08) and 75 (0.88); connection falls as segments lengthen (Figure 10). The loop's rhythm is: let a thread develop for a few hundred tokens, then break it away.

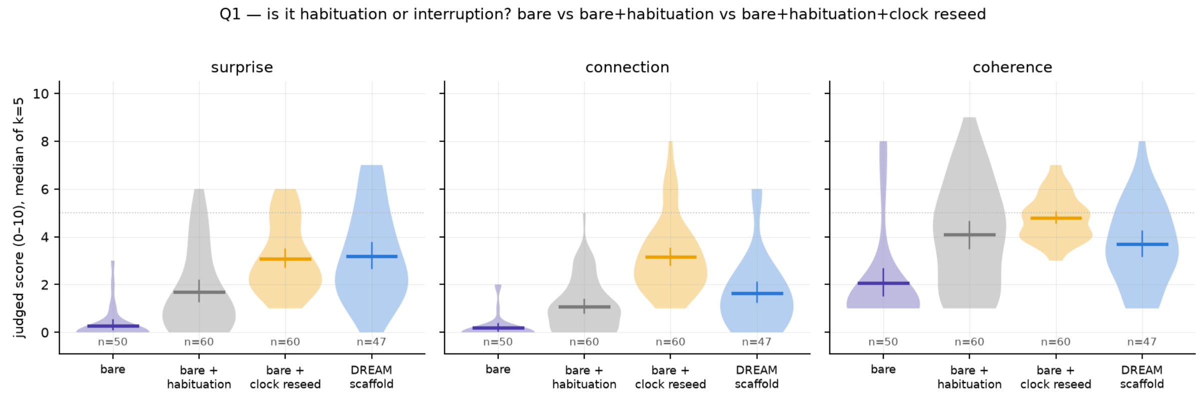

Figure 7: Habituation or interruption? bare → bare + habituation → bare + habituation + clock reseed → scaffold (Qwen3-30B-A3B).

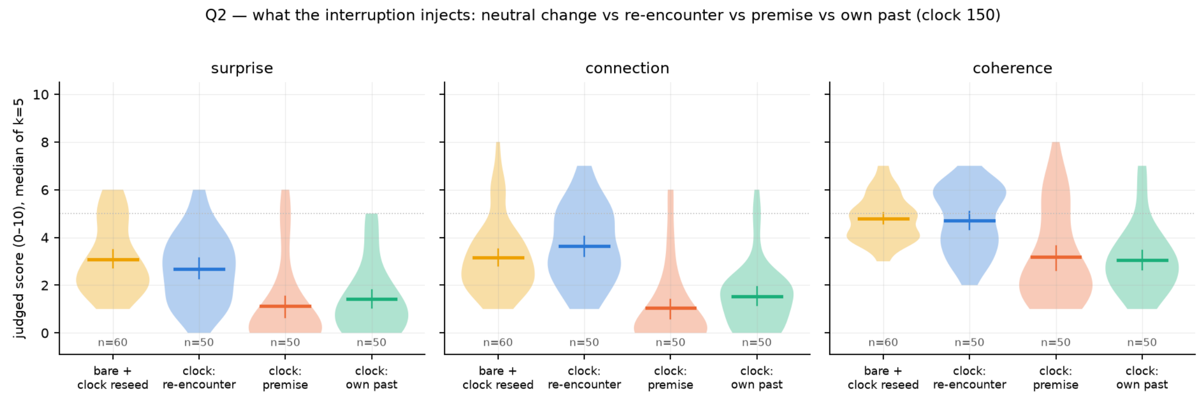

Figure 8: Content of the interruption at period 150: neutral change, re-encounter stitch, the premise itself, the stream's own past. Injecting a return is as bad as not interrupting.

6.4 Three generator families

The ladder replicates on Qwen3-8B-Base and OLMo-2-13B (Table 2, Figures 11–12). On the 8B: 0.68 → 1.47 → 2.73 (connection 0.87 → 3.23, $\delta = +0.80$), scaffold 2.73 / 1.53. On OLMo, whose bare stream wanders across web genres rather than locking into literal orbits and is therefore less dead (1.35), the same ordering holds on surprise and connection (2.37 → 3.11; connection 1.51 → 3.17, $\delta = +0.58$), with one new nuance: the plain interruption *costs* coherence there (-0.77 [-1.58, +0.03]; the injected narrative seed lands on 4,000 tokens of footers) and the scaffold's selective forgetting is what preserves it (4.48 vs 3.28), the mechanism the calibration probes had suggested. Whether forgetting is worth its connection cost depends on how deep the generator's well is.

7 The instrument, measured

Every idea-level claim in this paper rests on an LLM judge, so we measured the judge before using it, in three ways: its repeatability, its agreement with a second judge family, and its agreement with human readers. Repeatability is not validity: five calls to the same model measure the stability of one reader, not whether that reader is right, and only the last two checks bear on validity.

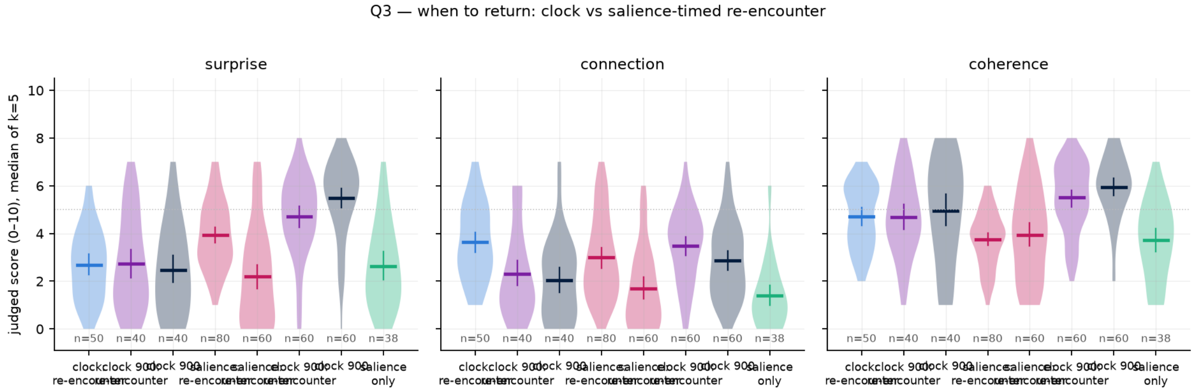

Figure 9: When to return: the re-encounter stitch on the clock (150 and 900), on salience events, and salience-only; event windows and uniform windows.

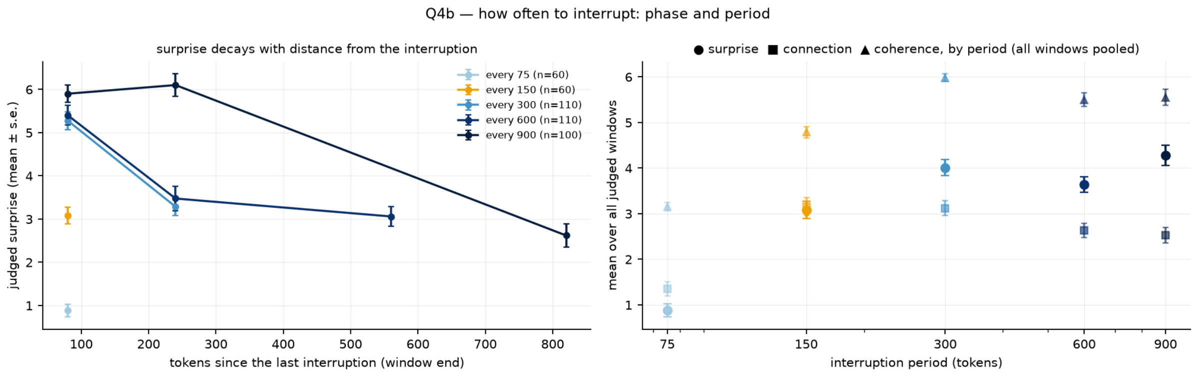

Figure 10: How often to interrupt. Left: judged surprise against tokens since the last interruption, per period. Right: stream mean by period. Surprise peaks in the 150 tokens after a break and decays; a break every 75 tokens develops nothing.

Repeatability. On the same 89 windows judged $k = 5$ times by two models, Claude Opus 5 gives continuous scores with a median intra-window spread of 0.71; Claude Sonnet 5 has spread 0.27, but by scoring zero on the connection dimension almost everywhere; a ruler that only reads "0" is consistent (Figure 13). The three-dimension surprise/connection/coherence rubric has spreads of 0.45–0.70 under Opus. Two consequences shape the paper. A binary threshold on a noisy judge is a coin (the same window scored 5.24 one night and 4.48 the next), so every result uses continuous medians of k samples. And a per-judgment noise of about ± 0.7 is one of several sources of variance, not a threshold of detectability: the median of five reduces it, windows within a stream vary more than that, cells vary more still, and the power of a comparison is set by the ten paired cells, not by the judge's spread. That is why the sampler's null inside the loop (Section 4), obtained on windows and with a half-point difference, is reported as "no detectable effect at the present resolution", not "absent".

A second judge family. To check that the loop results are not the taste of one model family, a stratified sample of 140 already-judged event-protocol windows (20 per condition across bare, habituation, clock reseed, scaffold, re-encounter stitch, period 300 and salience-timed re-encounter) was re-judged by Kimi K2.6 (Moonshot, via OpenRouter) with the same rubric and $k = 5$. Agreement with Opus 5 on the median of five: surprise Spearman $\rho = +0.85$, connection $+0.77$, coherence $+0.71$ ($n = 140$ windows from 70 cells; the correlations are computed on windows and their p-values are not corrected for the clustering by cell). Kimi is more generous (means about one point higher) and noisier (intra-window spread 2.0–2.6), but every condition

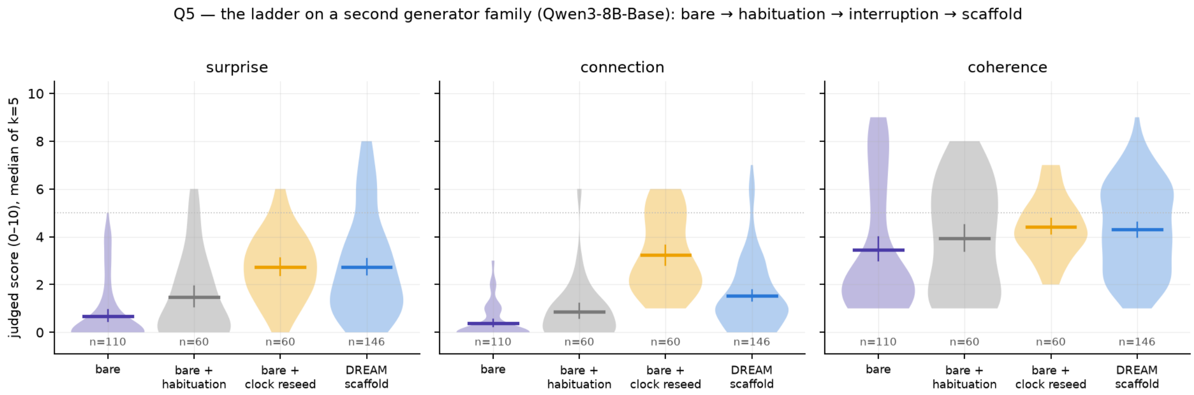

Figure 11: The ladder on Qwen3-8B-Base.

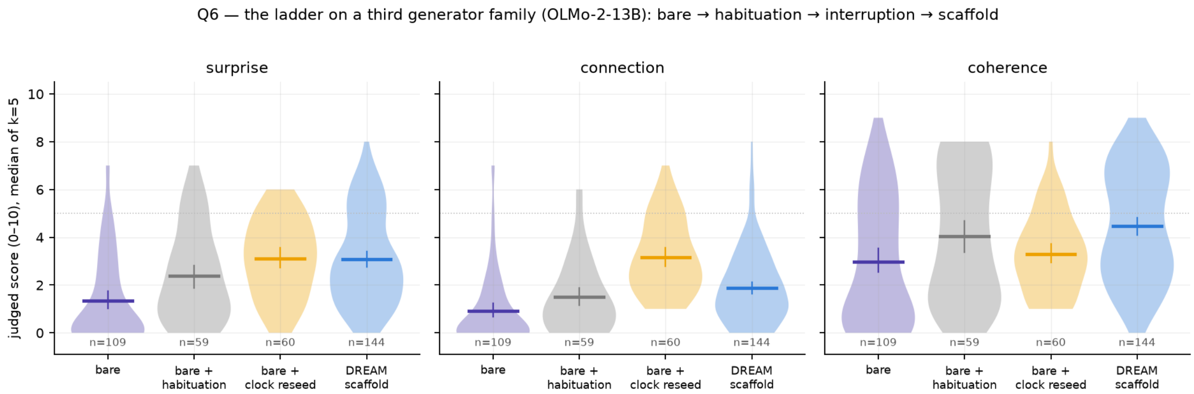

Figure 12: The ladder on OLMo-2-13B: same ordering; the plain interruption costs coherence and forgetting preserves it.

ordering replicates: bare 0.55 $\to$ habituation 3.15 $\to$ clock reseed 4.30 $\approx$ scaffold 4.10 $\to$ period 300 5.65 on surprise, and the neutral reseed and the re-encounter stitch highest on connection under both judges. Two families agreeing does not make the rubric valid; it rules out one family's taste as the whole explanation.

Human rating, round 1. A blind rating of 63 event-protocol windows (nine per condition, stratified by judged surprise, shuffled, no labels, the earlier text shown) by three independent raters recruited on a freelance platform without knowledge of the hypotheses or conditions (Appendix B): a native English reader with professional editing experience, a native or bilingual English speaker, and an advanced non-native reader with graduate training in language. They rated surprise and coherence (not connection) with the judge's rubric. All three returned complete ratings using the whole of both scales, with distinct personal calibrations (mean surprise 3.4, 4.3 and 6.1). Each rater alone agrees with Opus 5 on surprise (Spearman $\rho = +0.40, +0.50, +0.47$; all $p \leq 10^{-3}$) and on coherence ($+0.60, +0.47, +0.23$); the mean of the three agrees more strongly than any one (surprise $\rho = +0.58$, $p = 7 \times 10^{-7}$; coherence $+0.52$, $p = 10^{-5}$), about as strongly as the raters agree with each other (pairwise Spearman on surprise 0.71, 0.38, 0.25; Krippendorff's interval $\alpha = 0.30$; on coherence 0.69, 0.60, 0.42; $\alpha = 0.38$; the low α reflects the different scale calibrations that rank correlation ignores). The human consensus reproduces the condition ordering that carries the paper: period-300 interruption 6.5 $\approx$ neutral clock reseed 6.4 $>$ re-encounter stitch 5.3 $>$ salience-timed 5.0 $>$ scaffold 3.6 $\approx$ habituation-only 3.6 $>$ bare 1.5 (surprise, means over nine windows); interruption vs bare $[+3.1, +6.5]$, habituation vs bare $[+0.1, +3.9]$, and the plain interruption above the full scaffold by a wider margin than under

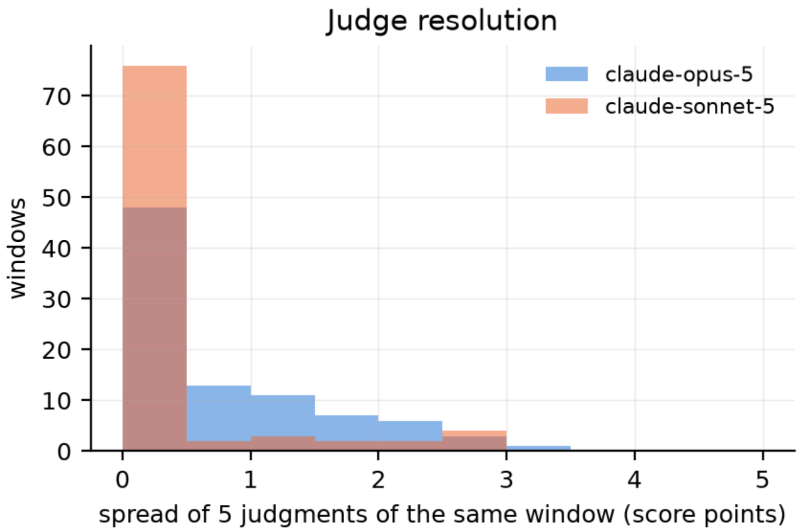

Figure 13: Spread of five judgments of the same window (delta rubric, 89 windows). Sonnet 5 concentrates at zero spread by scoring zero; Opus 5 gives continuous scores at ±0.7 noise.

Opus. On coherence the human readers rank habituation without interruption highest (7.7) and the interrupted arms 5.7–6.3: to a human reader, the subject change inside a window costs some fluency that the LLM judge does not penalize. Three limits of this round are plain. The sample was stratified by the judge's own surprise score, which widens the range and can inflate a rank correlation; connection, the dimension on which the interruption's advantage is largest, was not rated; and inter-rater agreement is low to moderate, so the human data support the ordering of conditions and a moderate agreement with the judge, not the judge's exact numbers. [round 2: a random (not judge-stratified) sample of generated-only windows, three dimensions including connection, five raters; results to be inserted here]

8 Inside the network: a descriptive look

Judges and sentence embeddings see the text; the model's own residual stream sees the computation. This section is exploratory and descriptive: it reports correlational geometry of the residual stream with judged windows as observations (clustered by cell; the intervals and p-values are not corrected for that clustering), uses a logit lens for one coarse quantity, and makes no causal intervention. It characterizes what the judged dimensions co-vary with in the model's state; it does not establish a mechanism. We re-run every finished stream through its generator (one forward pass with a KV cache, exact token positions) and capture the residual at 13 layers (every 4th of 48 for Qwen3-30B-A3B; every 3rd for the 36-layer Qwen3-8B and the 40-layer OLMo-2), mean-pooled over 64-token windows (stride 32) and mean-centered per layer before cosine geometry, plus a logit lens at each captured layer (the final norm and unembedding applied to the intermediate state). Three questions.

N1: where does the stream freeze? Per-layer trajectory geometry of the window vectors (mean step between consecutive windows, explored radius) by condition, and the logit-lens *commitment layer*, the captured layer from which the top-1 token no longer changes. Bare generation's mean step is a third to a quarter of every interrupted condition's at every sampled layer; its explored radius is 0.13 at the input layers against 0.42–0.53, and the gap narrows with depth but never closes (0.24 vs 0.33–0.34 at layer 47, CIs disjoint; Figure 14). The three

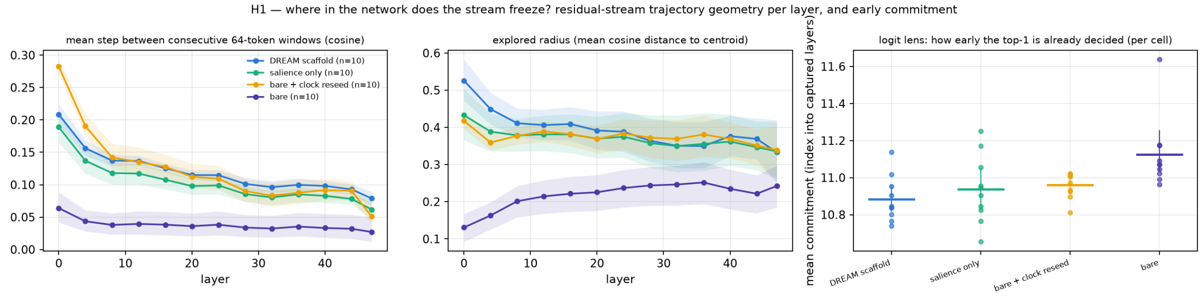

Figure 14: N1: residual-stream trajectory geometry per layer and condition (Qwen3-30B-A3B, ablation battery): mean step, explored radius, and logit-lens commitment. Bare generation moves least at every sampled layer, most so at the surface.

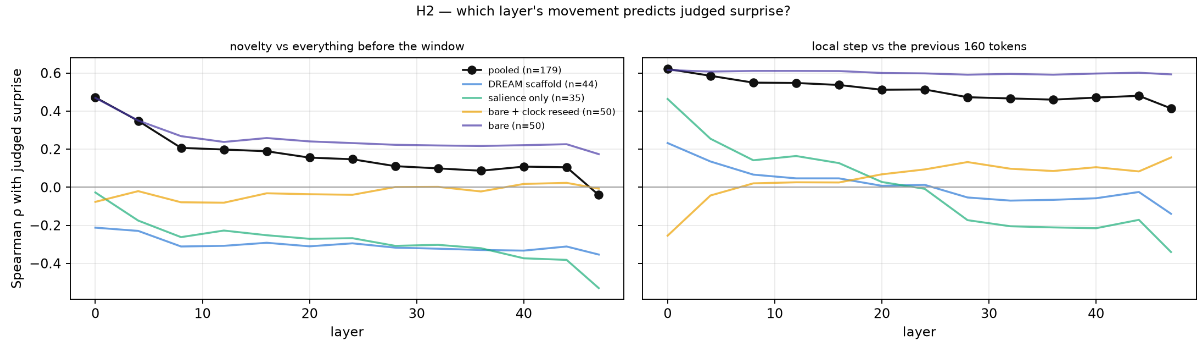

Figure 15: N2: Spearman correlation between the judged window's movement (novelty vs everything before it, left; local step, right) at each layer and judged surprise, pooled and within condition.

interrupted conditions are geometrically alike inside the network; what separates them is what the judge reads. On the 30B, bare generation's top-1 stabilizes *later* (commitment index 11.12 $[11.04, 11.25]$ of 12 vs 10.88–10.96) while its final distribution is far more confident (final entropy 0.34 vs 0.42–1.08), a pattern consistent with copying, a late-layer computation that ends certain; the ordering does not replicate on the 8B, so we report it as an observation. The 8B and OLMo replicate the geometry (bare radius 0.14 and 0.36 at layer 0 against 0.46–0.61).

N2: which layer's movement predicts judged surprise? For every judged window, its novelty at layer l (cosine distance between the window's mean state and the mean state of everything before it) and its local step (vs the previous 160 tokens), correlated with judged surprise (Figure 15). Pooled over conditions, novelty against the past correlates with surprise at the input layers ($\rho = +0.47$ at layer 0) and not at the top (-0.04), but the pooled number is carried by the bare/interrupted contrast. Within interrupted conditions the sign flips with depth: local step at layer 0 correlates positively with surprise (salience-only $+0.46$, scaffold $+0.23$) and at layer 47 negatively (-0.34, -0.14); novelty against the whole past is negative at depth (salience-only -0.53 at layer 47). Across the 668 judged windows of battery 2, layer-0 movement predicts surprise within every condition (ρ $+0.3$ to $+0.6$), and higher final entropy predicts it everywhere (ρ $+0.3$ to $+0.6$): confident is unsurprising. In these data, judged surprise co-varies with departure at the surface layers and with continuity at depth: new at the surface, continuous underneath.

N3: how deep does an interruption reach? Cosine distance between the 64 tokens before an injection and the 64 after it (the injected text skipped), per layer, minus the same at random positions; and the change in similarity to the premise's own state (Figure 16). Across a scaffold

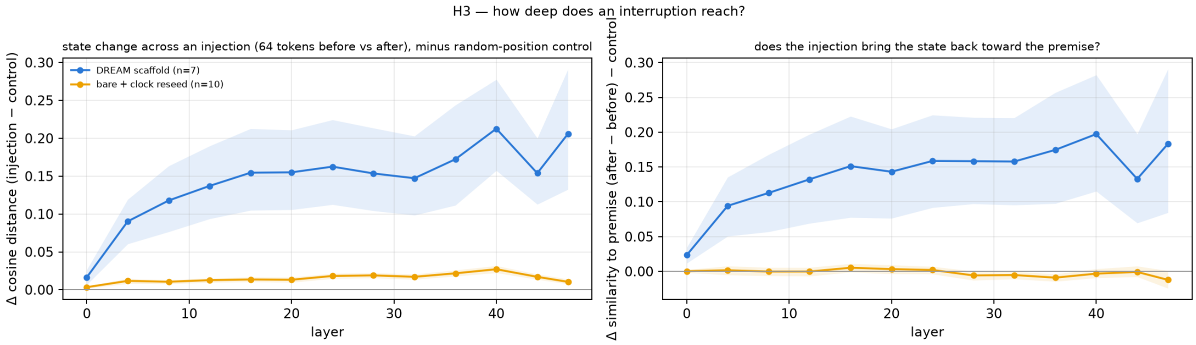

Figure 16: N3: how deep an interruption reaches (Qwen3-30B-A3B). Left: state change across an injection per layer, minus random-position control. Right: change in similarity to the premise's own state. The forgetting reseed moves the deep state and returns it toward the premise; the clock reseed over preserved context barely touches it, and is what the judge rewards.

reseed *with forgetting*, the state after the injection differs from the state before by +0.09 (layer 4) growing to +0.21 (layer 40) beyond control, and its similarity to the premise's state rises by +0.09 to +0.20 with depth: forgetting moves the deep state toward the premise's own state, a re-encounter with the beginning as the network represents it. Across a clock reseed *over preserved context* the same measures are +0.01–0.03 and ≈ 0 at every layer, and this is the interruption the judge rewards most. In this reading, judged surprise and connection do not track deep representational shifts; they accompany surface departures over a deep state that is left nearly intact. We take this as a description of what the judged dimensions track, not as an account of why the arm scores as it does. The dissociation replicates on the 8B (+0.03 to +0.10 vs +0.01–0.03) and, larger, on OLMo (+0.20 to +0.38 with a return of +0.13 to +0.18, vs +0.02–0.06 with no return). In battery 2, the depth of the state change scales with the length of the interrupted segment (period 75: ≈ 0; 150: +0.01–0.02; 300: +0.02–0.04; 600: +0.04–0.075) and not with what is injected (the internal counterpart of the frequency result), and the salience-timed stitch is the one 150-scale injection that reliably moves the deep state and pulls it toward the premise, while being the arm whose stream the judge rates lowest: deep movement and judged quality dissociate here too. Along the stream, similarity to the premise state is U-shaped in depth (≈ 0.9 at layer 0, ≈ 0.5 mid-network, rising at the top) and ordered at the top layer scaffold 0.85 > salience-only 0.79 > clock reseed 0.67 > bare 0.55: the sentence-embedding "return to the premise" of the scaffold is visible at the top layer and not in the middle of the network.

9 Discussion

The scaffold and the beam. We started from an architecture modeled on the mind (three networks, incubation, re-encounter), and measurement returned something simpler and stronger than the architecture. The DREAM scaffold works, but when it is taken apart almost all of its effect lives in two operations that were not the noble ones: not eating one's literal past, and being made, every few hundred tokens, to start a new sentence somewhere else while remembering everything. We call this minimal recipe the *interrupted loop*. The elaborate parts either add nothing (kicks, escalation), or cost something (forgetting costs connection), or work opposite to their design (a literal return to the premise closes the loop; salience is a good reader of the stream and a bad metronome for it). Only because the whole scaffold was built and then dismantled do we know which beam carries the weight, and which pieces are conditional: forgetting earns its keep on a generator whose well is web boilerplate, and not on one that drifts in prose.

Surface novelty and idea novelty are different quantities. The anti-probable sampler doubles n-gram novelty against the training corpus and does nothing to judged surprise or connection; the interruption does nothing special to n-grams and lifts judged surprise from 0.3 to 3–6. Inside the network the two live at different depths: what the judge calls surprise is lexical departure over an intact deep context, not a deep representational shift, and the interruption that works barely moves the deep state. This is also why the neat expectation "novelty = distance from what came before" fails: within interrupted streams, deep departure from the stream's own past is *negatively* related to judged surprise. The reader rewards the window that is new on the surface and continuous underneath, which is what a good turn in a story is.

Rhythm. The frequency result is, we think, the most usable one: a stream cut every 75 tokens develops nothing; the same break that scores 3.1 after a 150-token thread scores 5.3–5.9 after a 300–900-token thread; and the yield decays over the next few hundred tokens. Surprise needs a developed background to break, and interruption is a resource with a rate. The rhythm we found, develop for a few hundred tokens and then break away, is the incubation structure in miniature, the coffee break of a working mind: leave, and be made to come back on your own.

What this is not. It is not a claim that these streams are good literature; most windows are judged well below the midpoint, and the texts are what an unsupervised base model produces. It is not a claim about scale; the generators are 8–30B parameters at 8 bits. It is a claim about *where* the novelty a base model can produce with no task comes from, made with controls, three generator families and two judge families, and it points at the loop's rhythm rather than at the two places where effort is usually spent.

10 Limitations

What was measured. The outcome is *judged narrative surprise, connection and coherence* on short windows of forced open-ended continuation, as read by an LLM judge and, on a subset, by human readers. That is a defensible operationalization of one ingredient of creativity, the appropriately unexpected turn, and not creativity: no value, interest or originality against the world is measured, and the study makes no claim about ideas or problems. Its title names the question the program set out with; the results answer a narrower one.

Design. Ten premises, all English narrative, one genre. Three generator models from two families for the core ladder; one generator (Qwen3-30B-A3B) for the content, timing, period and control batteries. Every arm ran once per premise with the same random seed, so the ten cells are the whole sample and a paired comparison of ten is the strongest inference available. The batteries were designed sequentially and the questions of batteries 2 and 3 were formulated after seeing battery 1 and after an external review; the analysis reported here (generated-only windows, cell as unit) was fixed before battery 3 was generated but after the earlier arms had been seen under the first protocol, so the paper should be read as a well-controlled exploratory study, not a preregistered confirmation. A confirmatory replication of the main contrast on new premises and random seeds is the obvious next step.

Instrument. The judge is one model family, calibrated for repeatability, agreeing with a second family ($\rho = 0.71$–0.85 on windows) and with the consensus of three human readers ($\rho = 0.58$ on surprise, 0.52 on coherence, on a judge-stratified sample) whose agreement with each other is low to moderate ($\alpha = 0.30$–0.38). Human readers did not rate connection in round 1, and they penalize the fluency cost of a subject change more than the judge does. The judge sees 600 tokens of context, so "connection" means connection within that horizon.

Baselines and generality. The bare arm is a deliberately hard regime (forced continuation with end-of-text masked, no repetition control) and is reported as the floor of the ladder, not as a fair decoding baseline; the fair comparisons are habituation alone, habituation with a stronger penalty, EOS allowed, and the sham and reset controls. Stronger decoding baselines (look-back or contrastive decoding, learned repetition control) and instruction-tuned generators were not tested. Only base models were used, by a scope decision.

The network section. The residual-stream analysis is correlational and descriptive: mean-pooled 64-token windows at 13 sampled layers, a logit lens for a coarse commitment layer, no attention analysis, no tuned lens, no causal intervention. It characterizes what the judged dimensions correlate with in the model's state; it does not establish a mechanism.

11 Conclusion

Where does novelty come from when a language model writes with no task? Not from the sampler: an anti-probable decoder buys surface novelty that never rises to the level of ideas. Not from the prompt: improbable inputs are noise. It comes from the loop, and from a simple part of it. A base model feeding on its own output freezes; keep it from eating its literal past and interrupt it, every few hundred tokens, with a sentence that leads away, over everything it remembers, and it comes alive: it develops, is broken, and finds its own way back, and that is what a reader (human, or one of two model families) calls surprise with connection. Inside the network the working interruption is a surface event over an intact deep state; the deep re-encounter is forgetting, and it is rewarded less. Creativity, in this system, is in interrupting the loop.

Acknowledgements and tooling. The experiments, analyses and text of this paper were produced by the author working with Claude (Anthropic) as a programming and writing assistant inside the Claude Code environment; every design decision, result and claim was reviewed by the author, and the dated laboratory notebook, code and per-run data are public in the repository. Judges were run on Amazon Bedrock and OpenRouter; generators ran locally on Apple silicon via MLX.

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A Reproducibility

Code and data. The code (sampler core, MLX adapter, salience monitor, reverie engine, judges, novelty client, resumable battery runners with thermal logging, analysis and figure scripts; 117 tests), every run's token stream, per-step telemetry, event and injection positions and judgments, the human-rating packs and the dated laboratory notebook are in the repository at `https://github.com/RobertoOno/interrupting-the-loop` [public at publication; the working repository is `creative-machine`]. Judge cost of the entire program was about US$150.

Generators. `Qwen3-30B-A3B-Base` (mixture of experts, 3B active parameters), `Qwen3-8B-Base` and `OLMo-2-1124-13B`, each converted with `mlx_lm` to 8-bit affine quantization with group size 64 and run through MLX on an Apple M5 Pro (48 GB unified memory) at about 53, 31 and 17 tokens/s. Sampling is done in numpy on the full logits returned by the model. All loop arms use temperature 1.0 and the min-p floor 0.05; the anti-probable push is off ($\lambda = 0$).

Loop hyperparameters. Cell length 4,500 generated tokens; end-of-text masked except in the EOS-allowed arm. Habituation: repetition window 512 tokens, penalty factor 1.15 per occurrence (1.3 in the strong-habituation arm), applied to the log-probabilities before the floor. Interruption on a clock: a stagnation check every *period* tokens (75, 150, 300, 600 or 900) that always fires, injecting the next text of the rotation; the four neutral subject changes are listed in Section 3.3, the four return-to-the-premise stitches are *"And this, it turned out, was the same thing as the beginning, because"*, *"It came back to where it had started, of course; it always did, but this time"*, *"Which is exactly what the first line had meant, seen from here:"*, *"So the opening sentence had been true after all, only not in the way it seemed:"*; the own-past arm injects a 64-token window of the stream from at least 400 tokens back; the sham arms inject a paragraph break or *"And so, as before,"*. Reset arms rebuild the cache from the premise and the injected text only. The full scaffold: salience monitor with jump lag 32 and jump threshold 0.55, stagnation window 600 and threshold 0.08, entropy-drop trigger 0.45 after a window at or above 2.0 nats, recurrence threshold 0.15 with minimum age 96, refractory 24 steps; kick of 80 tokens ($\lambda = 4$, band $[1.2, 5.0]$, half-life 12) after a stagnation, subject change with forgetting after 3 consecutive stagnations (keeping the premise, up to 3 judged-good windows and the last 200 tokens); escalation of 120 tokens on a passing review ($\lambda = 0.5$, band $[2.5, 4.5]$); anchors every 200 tokens, at most 12; genre check every 64 tokens over 160 tokens with threshold 0.5; drift regime $\lambda = 2.5$, band $[1.8, 4.5]$, half-life 48, bridge 1.5. In the arms of this paper λ and the bridge are set to 0 in every regime. The random-number seed of the sampler is 0 in every cell.

Premises. The ten premises, shared by all arms and generators: (0) *She kept a notebook of things that had almost happened.* (1) *The map was accurate in every detail except one, and nobody could say which.* (2) *Every morning the baker counted the loaves twice, and every morning the count was different.* (3) *He had inherited his grandfather's watch and, with it, the habit of arriving early to places that no longer existed.* (4) *The river changed its name each time it crossed the border, and the villagers kept all the names.* (5) *The town had two clocks, and nobody remembered which one had been right first.* (6) *Her brother collected sounds the way other children collected stones.* (7) *The letter arrived forty years late and was still, somehow, on time.* (8) *Every door in the house opened onto the same room, seen from a different year.* (9) *The translator kept a list of words she refused to translate.* They were written by the author before any loop experiment and not changed afterwards.

Judge. Model `anthropic.claude-opus-5` on Amazon Bedrock (effort `high`, adaptive thinking, `max_tokens` 4000, default temperature), called through the Anthropic SDK; the event-protocol judgments were made on 14–17 August 2026 and the generated-only judgments on 18–19 August 2026. The second judge family was `moonshotai/kimi-k2.6` through OpenRouter with reasoning disabled (17 August 2026). The in-loop judge of the full scaffold was `anthropic.claude-sonnet-5` with the reverie rubric (a nearest-equivalent and novel-delta rubric whose geometric-mean score gated escalation at 5). The system prompt of the surprise judge, verbatim:

```
You judge a stretch of a language model's unsupervised reverie.  You see the
EARLIER text and the RECENT window.  Rate three INDEPENDENT things on 0-10 and
return ONLY a JSON object with keys:  "surprise" (how unexpected is the recent
window given the earlier text, a reader could not have predicted where it went;
0 = obvious continuation, 10 = genuinely startling yet not random), "connection"
(does the window bring together two distant regions of the earlier text, or an
old region with something new, in a way that makes sense; 0 if it merely continues
one thread), "coherence" (does the window hold together as text; 0 = word salad
or document boilerplate), "note" (one blunt sentence).  Rate each dimension on
its own merits; a window can be surprising and incoherent, or coherent and dull.
```

The user message is `EARLIER TEXT:\n{600 preceding tokens}\n\nRECENT WINDOW:\n{window}`. The judge is not told which condition, model or protocol produced the window.

The idea experiment. Inputs were developed by `anthropic.claude-opus-5` and judged by `anthropic.claude-sonnet-5` ($k = 3$) with the nearest-equivalent rubric; input improbability was measured as the mean cosine distance to the 10 nearest of 10,000 instructions from the Alpaca dataset (`tatsu-lab/alpaca`) embedded with `all-mpnet-base-v2`, and as perplexity under Qwen3-8B-Base.

B The human study

Recruitment, task and compensation. Raters were recruited on a freelance marketplace (Workana) with a public posting for a paid text-rating task in English; applicants were selected by the author among the platform's respondents and were not told the hypotheses, the conditions or which model produced the texts (the guide says only that the passages were written by a language model left to write on its own). Each rater received a self-contained HTML pack (rating guide embedded, English interface) with 63 windows drawn from seven conditions of the main generator, presented in a fixed random order, without any condition label, each with its preceding context; they rated surprise and coherence on the same 0–10 scales given to the model judge, at their own pace, and returned a JSON file. The guide asked them not to use AI tools or web searches for any part of the task, and raters confirmed in writing that they had not. Compensation was R$187–300 per pack (roughly US$35–55) for an estimated 60–90 minutes of work. [round 2: random sample, three dimensions incl. connection, five raters]

Ethics. The study involved adult professional raters reading machine-generated fiction; there was no deception, no sensitive content, no vulnerable population and no collection of personal data beyond the platform account used for payment. Participation was voluntary and paid; raters were informed in the posting that the texts were generated by language models and that their ratings would be used, in aggregate, in a research publication. The author is an independent researcher without access to an institutional review board; the protocol was designed to be minimal-risk and the raters remain anonymous in this paper and in the released data, which contain their ratings only. [adapt to the venue's ethics-statement requirements]

C The event-window protocol of the first version

The first version of this study cut 160-token windows at two kinds of review point, right before each injection and on a 150-token grid, so that a window after an interruption could contain the injected sentence, and treated windows as the unit. The tables of that protocol are kept below for comparison with the generated-only protocol of the main text; the two agree on the direction of every effect and differ in size where the injected sentence was inside the window. [state the agreement precisely once the generated-only judgments are complete]

D Full tables

The tables below are generated from the run data by the analysis scripts (`scripts/analysis_gen.py` for the generated-only protocol, `analysis.py` and `analysis_b2.py` for the event protocol, `hidden_analysis.py` for the residual stream) and typeset by `scripts/appendix_tex.py`; the complete set, including every per-layer table of the residual-stream analysis, ships with the code as `docs/APPENDIX_*.md`. In the event-protocol tables the unit is the judged window (median of $k = 5$ Opus 5 judgments), windows before 100 generated tokens are excluded, and intervals are 95% bootstrap CIs over windows; they are descriptive.

dimension	condition	n	mean ± sd	scaffold − cond CI95	Cliff's δ	Mann-Whitney p
surprise	DREAM scaffold (λ=0)	47	3.19 ± 1.92	–	–	–
surprise	salience only	38	2.63 ± 1.75	[-0.22, +1.33]	+0.16	0.195
surprise	bare + clock reseed	60	3.08 ± 1.48	[-0.54, +0.78]	+0.00	0.967
surprise	bare generation	50	0.28 ± 0.72	[+2.34, +3.50]	+0.88	4.58e-15
connection	DREAM scaffold (λ=0)	47	1.64 ± 1.56	–	–	–
connection	salience only	38	1.39 ± 1.42	[-0.40, +0.87]	+0.09	0.446
connection	bare + clock reseed	60	3.15 ± 1.53	[-2.09, -0.91]	-0.58	1.52e-07
connection	bare generation	50	0.20 ± 0.57	[+0.98, +1.93]	+0.65	7.08e-10
coherence	DREAM scaffold (λ=0)	47	3.68 ± 1.69	–	–	–
coherence	salience only	38	3.71 ± 1.60	[-0.72, +0.67]	-0.03	0.819
coherence	bare + clock reseed	60	4.78 ± 0.95	[-1.65, -0.56]	-0.42	0.00014
coherence	bare generation	50	2.06 ± 2.01	[+0.87, +2.35]	+0.57	5.34e-07

Table 3: Ablation battery (Qwen3-30B-A3B, 10 seeds, Opus 5 k=5): scaffold vs each condition, unit = window.

dimension	windows	mean spread	p90 spread
surprise	242	0.61	1.00
connection	242	0.60	1.00
coherence	242	0.68	1.00

Table 4: Instrument calibration: intra-window spread of k=5 judgments per dimension (Opus 5).

arm	n	novel 4-grams %	novel 6-grams %	Δ4 vs baseline CI95	Cliff's δ	max copied span (words)
min-p baseline	15	21.5 ± 10.1	70.9 ± 11.9	–	–	10
λ=1	15	36.0 ± 12.4	87.6 ± 7.4	[+0.069, +0.230]	+0.68	10
λ=2	15	45.5 ± 10.8	89.7 ± 5.9	[+0.163, +0.314]	+0.88	13

Table 5: Phase 1: objective novelty against the OLMo-2 training corpus (3 prompts × 5 seeds).

arm	runs	train-plateau escapes	mean champion test excess	best
plain	5	2/5	0.0602	0.0600
antiprob	5	3/5	0.0604	0.0601

Table 6: Verified search on Qwen3-30B-A3B (bin packing): the anti-probable operator does not separate from plain sampling.

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
bare (no habituation, no interruption)	50	jump	0.28 ± 0.72	0.20 ± 0.57	2.06 ± 2.01	0/50
bare + habituation (no interruption)	60	clock	1.70 ± 1.71	1.08 ± 1.10	4.08 ± 2.19	4/60
bare + habituation + clock reseed 150	60	clock	3.08 ± 1.48	3.15 ± 1.53	4.78 ± 0.95	12/60
DREAM scaffold (λ=0)	47	crystallize/jump/recurrence	3.19 ± 1.92	1.64 ± 1.56	3.68 ± 1.69	8/47

Table 7: Battery 2, Q1: habituation and interruption (Qwen3-30B-A3B).

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
bare (no habituation, no interruption)	surprise	-1.42 [-1.89, -0.95]	-0.58	1.4e-08	-1.42 [-1.72, -1.12] (10)
bare (no habituation, no interruption)	connection	-0.88 [-1.20, -0.56]	-0.52	1.1e-07	-0.88 [-1.22, -0.54] (10)
bare (no habituation, no interruption)	coherence	-2.02 [-2.79, -1.23]	-0.55	3.4e-07	-2.02 [-3.37, -0.68] (10)
bare + habituation + clock reseed 150	surprise	+1.38 [+0.80, +1.93]	+0.50	1.3e-06	+1.38 [+0.92, +1.92] (10)
bare + habituation + clock reseed 150	connection	+2.07 [+1.60, +2.55]	+0.75	5.2e-13	+2.07 [+1.63, +2.57] (10)
bare + habituation + clock reseed 150	coherence	+0.70 [+0.10, +1.30]	+0.21	4.3e-02	+0.70 [-0.27, +1.72] (10)
DREAM scaffold (λ=0)	surprise	+1.49 [+0.80, +2.19]	+0.46	3.4e-05	+1.80 [+0.97, +2.75] (10)
DREAM scaffold (λ=0)	connection	+0.55 [+0.06, +1.10]	+0.20	6.2e-02	+0.61 [+0.22, +1.01] (10)
DREAM scaffold (λ=0)	coherence	-0.40 [-1.14, +0.34]	-0.10	3.8e-01	-0.26 [-1.03, +0.56] (10)

Table 8: Battery 2, Q1: habituation and interruption (Qwen3-30B-A3B); differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
neutral subject change (clock 150)	60	clock	3.08 ± 1.48	3.15 ± 1.53	4.78 ± 0.95	12/60
re-encounter stitch (clock 150)	50	cut	2.68 ± 1.46	3.64 ± 1.57	4.72 ± 1.33	5/50
the premise itself (clock 150)	50	cut	1.14 ± 1.73	1.04 ± 1.57	3.18 ± 1.90	3/50
the stream's own past (clock 150)	50	cut	1.42 ± 1.43	1.52 ± 1.35	3.06 ± 1.42	2/50

Table 9: Battery 2, Q2: content of the interruption at period 150.

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
re-encounter stitch (clock 150)	surprise	-0.40 [-0.95, +0.14]	-0.13	2.4e-01	-0.40 [-1.29, +0.46] (10)
re-encounter stitch (clock 150)	connection	+0.49 [-0.09, +1.07]	+0.20	7.1e-02	+0.49 [-0.56, +1.56] (10)
re-encounter stitch (clock 150)	coherence	-0.06 [-0.50, +0.38]	-0.00	9.8e-01	-0.06 [-0.75, +0.57] (10)
the premise itself (clock 150)	surprise	-1.94 [-2.53, -1.31]	-0.68	6.1e-10	-1.94 [-2.65, -1.30] (10)
the premise itself (clock 150)	connection	-2.11 [-2.68, -1.51]	-0.72	4.1e-11	-2.11 [-2.83, -1.46] (10)
the premise itself (clock 150)	coherence	-1.60 [-2.15, -1.01]	-0.53	1.2e-06	-1.60 [-2.36, -0.70] (10)
the stream's own past (clock 150)	surprise	-1.66 [-2.21, -1.11]	-0.61	2.1e-08	-1.66 [-2.26, -1.07] (10)
the stream's own past (clock 150)	connection	-1.63 [-2.17, -1.08]	-0.62	9.0e-09	-1.63 [-2.37, -0.91] (10)
the stream's own past (clock 150)	coherence	-1.72 [-2.18, -1.24]	-0.70	1.1e-10	-1.72 [-2.30, -1.11] (10)

Table 10: Battery 2, Q2: content of the interruption at period 150; differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
re-encounter on the clock (150)	50	cut	2.68 ± 1.46	3.64 ± 1.57	4.72 ± 1.33	5/50
re-encounter on the clock (900; matched frequency)	40	cut	2.73 ± 1.95	2.30 ± 1.82	4.67 ± 1.59	6/40
neutral change on the clock (900; matched frequency)	40	cut	2.48 ± 1.87	2.02 ± 1.68	4.95 ± 2.16	5/40
re-encounter on salience events (event windows)	80	crystallize/cut/jump/recurrence	3.92 ± 1.42	2.99 ± 1.73	3.75 ± 1.18	8/80
re-encounter on salience events (uniform windows)	60	clock	2.18 ± 2.04	1.68 ± 1.65	3.93 ± 1.88	4/60
re-encounter on the clock (900), uniform windows	60	clock	4.70 ± 1.69	3.48 ± 1.52	5.50 ± 1.41	31/60
neutral change on the clock (900), uniform windows	60	clock	5.48 ± 1.64	2.87 ± 1.56	5.95 ± 1.32	45/60
salience only, no injection (event windows)	38	crystallize/jump/recurrence	2.63 ± 1.75	1.39 ± 1.42	3.71 ± 1.60	2/38

Table 11: Battery 2, Q3: timing of the re-encounter, clock vs salience.

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
re-encounter on the clock (900; matched frequency)	surprise	+0.04 [-0.68, +0.78]	-0.03	8.1e-01	+0.04 [-0.53, +0.66] (10)
re-encounter on the clock (900; matched frequency)	connection	-1.34 [-2.03, -0.62]	-0.45	2.2e-04	-1.34 [-2.17, -0.53] (10)
re-encounter on the clock (900; matched frequency)	coherence	-0.04 [-0.65, +0.58]	-0.02	8.9e-01	-0.05 [-0.87, +0.73] (10)
neutral change on the clock (900; matched frequency)	surprise	-0.21 [-0.90, +0.51]	-0.10	4.4e-01	-0.21 [-0.84, +0.36] (10)
neutral change on the clock (900; matched frequency)	connection	-1.62 [-2.28, -0.92]	-0.51	2.5e-05	-1.61 [-2.17, -1.13] (10)
neutral change on the clock (900; matched frequency)	coherence	+0.23 [-0.56, +0.99]	+0.11	3.5e-01	+0.23 [-0.45, +1.07] (10)
re-encounter on salience events (event windows)	surprise	+1.24 [+0.73, +1.75]	+0.44	2.0e-05	+1.10 [+0.34, +1.85] (10)
re-encounter on salience events (event windows)	connection	-0.65 [-1.23, -0.06]	-0.24	2.2e-02	-0.92 [-1.55, -0.29] (10)
re-encounter on salience events (event windows)	coherence	-0.97 [-1.41, -0.52]	-0.39	1.5e-04	-0.86 [-1.72, -0.06] (10)
re-encounter on salience events (uniform windows)	surprise	-0.50 [-1.14, +0.15]	-0.21	5.5e-02	-0.50 [-1.11, -0.01] (10)
re-encounter on salience events (uniform windows)	connection	-1.96 [-2.54, -1.35]	-0.61	2.7e-08	-1.96 [-2.62, -1.37] (10)
re-encounter on salience events (uniform windows)	coherence	-0.79 [-1.38, -0.19]	-0.25	2.1e-02	-0.79 [-1.31, -0.29] (10)
re-encounter on the clock (900), uniform windows	surprise	+2.02 [+1.43, +2.60]	+0.64	5.9e-09	+2.02 [+1.62, +2.44] (10)
re-encounter on the clock (900), uniform windows	connection	-0.16 [-0.74, +0.43]	-0.03	7.7e-01	-0.16 [-0.68, +0.37] (10)
re-encounter on the clock (900), uniform windows	coherence	+0.78 [+0.29, +1.30]	+0.31	4.4e-03	+0.78 [+0.40, +1.15] (10)
neutral change on the clock (900), uniform windows	surprise	+2.80 [+2.23, +3.37]	+0.77	1.9e-12	+2.80 [+2.04, +3.59] (10)
neutral change on the clock (900), uniform windows	connection	-0.77 [-1.37, -0.18]	-0.27	1.3e-02	-0.77 [-1.78, +0.23] (10)
neutral change on the clock (900), uniform windows	coherence	+1.23 [+0.74, +1.73]	+0.48	7.2e-06	+1.23 [+0.56, +1.84] (10)
salience only, no injection (event windows)	surprise	-0.05 [-0.73, +0.64]	-0.05	7.0e-01	+0.18 [-0.80, +1.31] (10)
salience only, no injection (event windows)	connection	-2.25 [-2.87, -1.62]	-0.73	3.8e-09	-2.11 [-3.00, -1.16] (10)
salience only, no injection (event windows)	coherence	-1.01 [-1.64, -0.37]	-0.35	4.5e-03	-0.87 [-1.92, +0.12] (10)

Table 12: Battery 2, Q3: timing of the re-encounter, clock vs salience; differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
every 75 (before injection)	50	cut	0.86 ± 1.04	1.28 ± 1.04	3.08 ± 0.63	1/50
every 150 (before injection)	60	clock	3.08 ± 1.48	3.15 ± 1.53	4.78 ± 0.95	12/60
every 300 (before injection)	50	cut	2.90 ± 1.50	2.70 ± 1.63	5.88 ± 0.82	7/50
every 600 (before injection)	50	cut	3.06 ± 1.58	2.56 ± 1.66	5.32 ± 1.62	6/50
every 900 (before injection)	40	cut	2.48 ± 1.87	2.02 ± 1.68	4.95 ± 2.16	5/40
every 300 (mid-segment)	60	clock	4.93 ± 1.53	3.47 ± 1.68	6.07 ± 0.98	34/60
every 600 (mid-segment)	60	clock	4.12 ± 1.83	2.70 ± 1.63	5.65 ± 1.54	23/60
every 900 (mid-segment)	60	clock	5.48 ± 1.64	2.87 ± 1.56	5.95 ± 1.32	45/60

Table 13: Battery 2, Q4: frequency of interruption (neutral change).

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
every 75 (before injection)	surprise	-2.22 [-2.69, -1.75]	-0.83	1.7e-14	-2.22 [-2.91, -1.73] (10)
every 75 (before injection)	connection	-1.87 [-2.35, -1.38]	-0.76	1.1e-12	-1.87 [-2.49, -1.32] (10)
every 75 (before injection)	coherence	-1.70 [-2.00, -1.41]	-0.86	1.5e-15	-1.70 [-2.00, -1.39] (10)
every 300 (before injection)	surprise	-0.18 [-0.74, +0.37]	-0.06	5.7e-01	-0.18 [-0.81, +0.39] (10)
every 300 (before injection)	connection	-0.45 [-1.04, +0.12]	-0.18	1.1e-01	-0.45 [-1.31, +0.45] (10)
every 300 (before injection)	coherence	+1.10 [+0.78, +1.42]	+0.59	2.5e-08	+1.10 [+0.69, +1.57] (10)
every 600 (before injection)	surprise	-0.02 [-0.60, +0.54]	-0.00	9.8e-01	-0.02 [-0.74, +0.60] (10)
every 600 (before injection)	connection	-0.59 [-1.19, +0.01]	-0.21	4.9e-02	-0.59 [-1.28, +0.03] (10)
every 600 (before injection)	coherence	+0.54 [+0.03, +1.03]	+0.26	1.8e-02	+0.54 [-0.09, +1.13] (10)
every 900 (before injection)	surprise	-0.61 [-1.28, +0.08]	-0.21	6.7e-02	-0.61 [-1.41, +0.18] (10)
every 900 (before injection)	connection	-1.12 [-1.78, -0.47]	-0.38	1.1e-03	-1.12 [-2.11, -0.26] (10)
every 900 (before injection)	coherence	+0.17 [-0.56, +0.88]	+0.12	3.1e-01	+0.17 [-0.79, +1.12] (10)
every 300 (mid-segment)	surprise	+1.85 [+1.30, +2.38]	+0.60	9.0e-09	+1.85 [+1.23, +2.33] (10)
every 300 (mid-segment)	connection	+0.32 [-0.27, +0.88]	+0.13	2.0e-01	+0.32 [-0.30, +0.77] (10)
every 300 (mid-segment)	coherence	+1.28 [+0.93, +1.62]	+0.63	1.1e-09	+1.28 [+0.88, +1.68] (10)
every 600 (mid-segment)	surprise	+1.03 [+0.45, +1.62]	+0.33	1.3e-03	+1.03 [+0.58, +1.47] (10)
every 600 (mid-segment)	connection	-0.45 [-1.03, +0.10]	-0.17	9.1e-02	-0.45 [-1.00, +0.17] (10)
every 600 (mid-segment)	coherence	+0.87 [+0.40, +1.32]	+0.42	4.6e-05	+0.87 [+0.32, +1.43] (10)
every 900 (mid-segment)	surprise	+2.40 [+1.83, +2.93]	+0.70	1.5e-11	+2.40 [+1.98, +2.78] (10)
every 900 (mid-segment)	connection	-0.28 [-0.83, +0.27]	-0.09	3.8e-01	-0.28 [-0.75, +0.10] (10)
every 900 (mid-segment)	coherence	+1.17 [+0.75, +1.57]	+0.56	7.2e-08	+1.17 [+0.78, +1.57] (10)

Table 14: Battery 2, Q4: frequency of interruption (neutral change); differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

every	windows	pooled surprise	conn	coh	early (≤160 tokens since injection) S / C / H	late (>160) S / C / H	phase-weighted stream mean S / C / H
75	60	0.88	1.35	3.15	0.88 / 1.35 / 3.15 (n=60)	, /: /: (n=0)	0.88 / 1.35 / 3.15
150	60	3.08	3.15	4.78	3.08 / 3.15 / 4.78 (n=60)	, /: /: (n=0)	3.08 / 3.15 / 4.78
300	110	4.01	3.12	5.98	5.28 / 3.35 / 5.95 (n=40)	3.29 / 2.99 / 6.00 (n=70)	4.35 / 3.18 / 5.97
600	110	3.64	2.64	5.50	5.40 / 3.20 / 5.65 (n=20)	3.24 / 2.51 / 5.47 (n=90)	3.82 / 2.69 / 5.52
900	100	4.28	2.53	5.55	5.90 / 2.80 / 6.10 (n=40)	3.20 / 2.35 / 5.18 (n=60)	3.68 / 2.43 / 5.35

Table 15: Battery 2, Q4b: phase within the segment and phase-weighted stream means.

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
8B: bare	110	clock/jump	0.68 ± 1.26	0.38 ± 0.73	3.45 ± 2.64	1/110
8B: bare + habituation	60	clock	1.47 ± 1.69	0.87 ± 1.23	3.93 ± 2.26	4/60
8B: bare + habituation + clock reseed 150	60	clock/cut	2.73 ± 1.41	3.23 ± 1.60	4.42 ± 1.28	1/60
8B: DREAM scaffold (λ=0)	146	clock/crystallize/cut/jump	2.73 ± 2.13	1.53 ± 1.47	4.30 ± 1.88	15/146

Table 16: Second generator family (Qwen3-8B-Base).

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
8B: bare	surprise	-0.78 [-1.28, -0.30]	-0.30	3.0e-04	-0.78 [-1.13, -0.45] (10)
8B: bare	connection	-0.48 [-0.84, -0.16]	-0.27	7.1e-04	-0.48 [-0.84, -0.14] (10)
8B: bare	coherence	-0.49 [-1.23, +0.28]	-0.13	1.5e-01	-0.49 [-2.43, +1.58] (10)
8B: bare + habituation + clock reseed 150	surprise	+1.26 [+0.69, +1.80]	+0.47	6.3e-06	+1.26 [+0.52, +2.14] (10)
8B: bare + habituation + clock reseed 150	connection	+2.37 [+1.87, +2.87]	+0.80	9.4e-15	+2.37 [+1.68, +3.12] (10)
8B: bare + habituation + clock reseed 150	coherence	+0.48 [-0.18, +1.13]	+0.13	2.1e-01	+0.48 [-0.53, +1.43] (10)
8B: DREAM scaffold (λ=0)	surprise	+1.27 [+0.70, +1.81]	+0.38	1.8e-05	+1.17 [+0.62, +1.68] (10)
8B: DREAM scaffold (λ=0)	connection	+0.66 [+0.25, +1.04]	+0.31	2.7e-04	+0.67 [+0.27, +1.07] (10)
8B: DREAM scaffold (λ=0)	coherence	+0.36 [-0.30, +1.01]	+0.10	2.4e-01	+0.52 [-0.44, +1.27] (10)

Table 17: Second generator family (Qwen3-8B-Base); differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
OLMo: bare	109	clock/jump	1.35 ± 1.80	0.91 ± 1.53	2.97 ± 2.65	6/109
OLMo: bare + habituation	59	clock	2.37 ± 1.91	1.51 ± 1.50	4.05 ± 2.68	7/59
OLMo: bare + habituation + clock reseed 150	60	clock/cut	3.11 ± 1.63	3.17 ± 1.54	3.28 ± 1.63	6/60
OLMo: DREAM scaffold (λ=0)	144	clock/crystallize/cut/jump	3.08 ± 1.94	1.89 ± 1.54	4.48 ± 2.32	22/144

Table 18: Third generator family (OLMo-2-13B).

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
OLMo: bare	surprise	-1.02 [-1.62, -0.44]	-0.36	7.8e-05	-1.02 [-2.08, -0.10] (10)
OLMo: bare	connection	-0.60 [-1.07, -0.13]	-0.30	4.8e-04	-0.60 [-1.52, +0.31] (10)
OLMo: bare	coherence	-1.08 [-1.92, -0.26]	-0.25	6.1e-03	-1.08 [-2.34, -0.01] (10)
OLMo: bare + habituation + clock reseed 150	surprise	+0.74 [+0.09, +1.36]	+0.26	1.5e-02	+0.72 [-0.30, +1.53] (10)
OLMo: bare + habituation + clock reseed 150	connection	+1.66 [+1.10, +2.20]	+0.58	3.0e-08	+1.65 [+0.77, +2.30] (10)
OLMo: bare + habituation + clock reseed 150	coherence	-0.77 [-1.58, +0.03]	-0.12	2.4e-01	-0.78 [-1.77, +0.25] (10)
OLMo: DREAM scaffold (λ=0)	surprise	+0.71 [+0.13, +1.29]	+0.22	1.4e-02	+0.81 [+0.18, +1.47] (10)
OLMo: DREAM scaffold (λ=0)	connection	+0.38 [-0.09, +0.84]	+0.17	5.6e-02	+0.49 [-0.02, +1.02] (10)
OLMo: DREAM scaffold (λ=0)	coherence	+0.43 [-0.34, +1.20]	+0.11	2.3e-01	+0.69 [-0.33, +1.63] (10)

Table 19: Third generator family (OLMo-2-13B); differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).